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DESIGN MILESTONES

AI SUPPORTED DESIGN PROCESS

SAFE TEMPERATURE DRINK PREPARATION SYSTEM FOR 1-7 YEAR OLD CHILDRENS

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KEY DESIGN MILESTONES

- **INITIAL PROBLEM RECOGNITION;**

CAREGIVERS STRUGGLE TO ESTIMATE DRINK TEMPERATURE SAFELY.

- **RESEARCH PHASE;**

OBSERVED NEED FOR ACCURACY, SAFETY, PORTABILITY, AND EASE OF USE.

- **INSIGHT DEFINITION**;

THE REAL ISSUE IS NOT ONLY HEATING, BUT GIVING DRINKS AT A CHILD-SAFE TEMPERATURE CONSISTENTLY.

- **PROBLEM FRAMING;**

HOW MIGHT I HELP CAREGIVERS PREPARE DRINKS FOR BABIES AND YOUNG CHILDREN WITHOUT RELYING ON GUESSWORK?

- **AI-SUPPORTED IDEATION;**

EXPLORATION OF FORM, MECHANISM, PORTABILITY, AND HEATING APPROACHES.

- **CONCEPT DEVELOPMENT;**

INTEGRATION OF HEATING SYSTEM, PROPELLER MIXING LOGIC, AND TEMPERATURE SENSING.

- **PROTOTYPE & EVALUATION**;

FORM BECAME BULKIER THAN EXPECTED; HANDLE AND USABILITY ISSUES APPEARED.

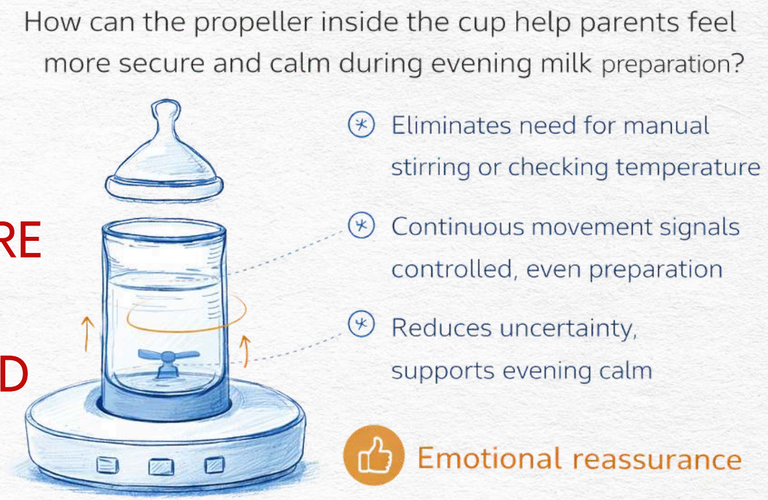
- **REFINEMENT;**

SHIFT TOWARD A SAFER, MORE PRACTICAL, MORE CONTROLLED SYSTEM.

PROJECT PROBLEM & GOAL

PROBLEM ;

CAREGIVERS OFTEN STRUGGLE TO PREPARE DRINKS AT THE CORRECT TEMPERATURE FOR BABIES AND YOUNG CHILDREN. INCORRECT TEMPERATURE CAN CAUSE BURNS OR DISCOMFORT. THE GOAL OF THIS PROJECT IS TO DESIGN A SAFE AND EASY SYSTEM THAT PREPARES DRINKS AT A CONTROLLED TEMPERATURE.

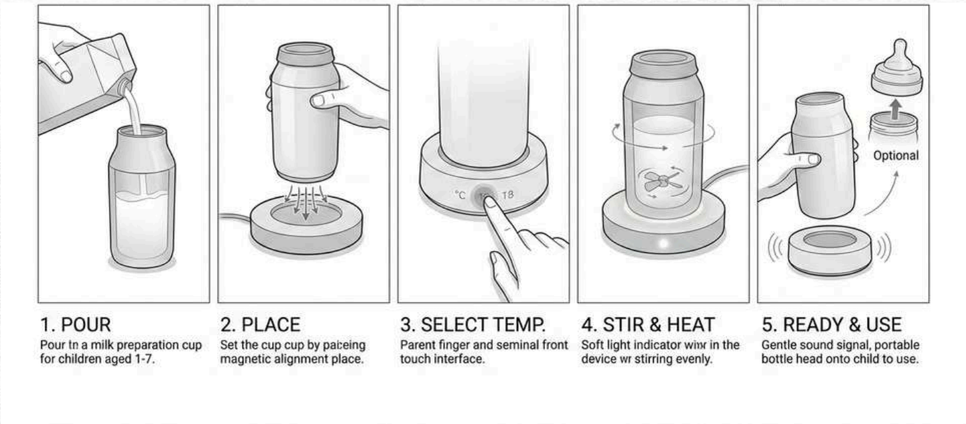


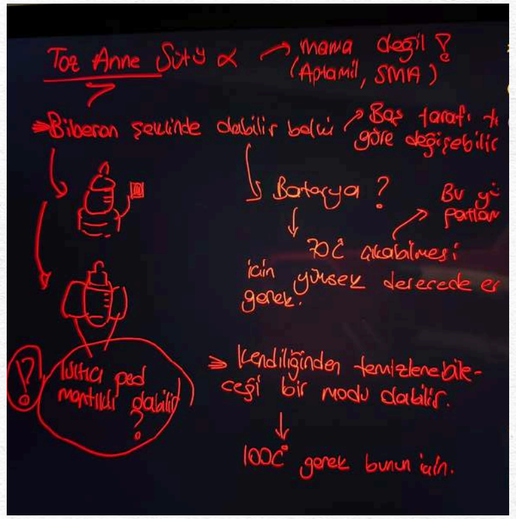
GOAL ;

DESIGN A SYSTEM THAT PREPARES DRINKS AT A SAFE TEMPERATURE QUICKLY AND EVENLY.

KEYWORDS :

- BABY SAFETY
- TEMPERATURE CONTROL
- CAREGIVER STRESS
- NIGHT PREPARATION
- SAFE DRINKING TEMPERATURE





RESEARCH INSIGHTS

INSIGHT 1
TEMPERATURE UNCERTAINTY ;

CAREGIVERS CANNOT ACCURATELY ESTIMATE DRINK TEMPERATURE.

MANY PARENTS TEST DRINK TEMPERATURE USING THEIR WRIST OR BY GUESSING.
THIS METHOD IS UNRELIABLE AND MAY RESULT IN DRINKS THAT ARE TOO HOT FOR BABIES.

INSIGHT 2
SENSITIVE MOUTHS ;

BABIES ARE VERY SENSITIVE TO HIGH TEMPERATURES.

EVEN SLIGHTLY OVERHEATED LIQUIDS CAN BURN A BABY'S MOUTH OR CAUSE DISCOMFORT.

INSIGHT 3
NIGHT-TIME STRESS ;

DRINK PREPARATION AT NIGHT INCREASES CAREGIVER STRESS.

WHEN BABIES WAKE UP AT NIGHT, CAREGIVERS NEED TO PREPARE DRINKS QUICKLY.
UNCERTAINTY ABOUT TEMPERATURE INCREASES ANXIETY.

INSIGHT 4
NEED FOR EVEN HEATING;

LIQUIDS MUST BE HEATED EVENLY.

IF HEAT IS NOT DISTRIBUTED PROPERLY, SOME PARTS OF THE DRINK CAN REMAIN TOO HOT.

OPPORTUNITY

DESIGN A SYSTEM THAT HEATS LIQUIDS EVENLY AND CONTROLS THE TEMPERATURE PRECISELY TO ENSURE SAFE DRINKING CONDITIONS FOR BABIES.

AI IDEATION

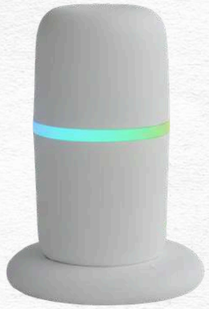
AI IDEATION → CONCEPT DEVELOPMENT → PROTOTYPE

PROBLEM CLARIFICATION

PROMPT;
"HELP ME DEFINE THE MAIN PROBLEM IN MY BABY DRINK PREPARATION PROJECT."
AI ROLE;
CLARIFIER
AI RESPONSE SUMMARY;
THE RESPONSE HELPED NARROW THE PROJECT FROM A GENERAL HEATING DEVICE TO A SYSTEM FOCUSED ON SAFE DRINK TEMPERATURE FOR BABIES AND YOUNG CHILDREN.
MY DESIGN TAKEAWAY;
I REALIZED THE REAL ISSUE WAS NOT HEATING DRINKS, BUT MAKING SURE THEY ARE SAFE FOR CHILDREN TO DRINK.

EXPANDING DESIGN POSSIBILITIES

PROMPT;
"GENERATE POSSIBLE PRODUCT DIRECTIONS FOR A PORTABLE BABY DRINK HEATING SYSTEM."
AI ROLE;
EXPANDER
AI RESPONSE SUMMARY;
THE AI SUGGESTED DIFFERENT PRODUCT FORMS, USAGE SCENARIOS, AND HEATING APPROACHES.
MY DESIGN TAKEAWAY;
THIS HELPED ME EXPLORE DIFFERENT PRODUCT SCALES AND UNDERSTAND THE IMPORTANCE OF PORTABILITY.



CONCEPT DEVELOPMENT

EARLY IDEA;

THE PROJECT INITIALLY STARTED AS A DRINK HEATING CONCEPT MAINLY FOCUSED ON WARMING MILK FOR BABIES AND CHILDREN.
AT THIS STAGE, THE FOCUS WAS MOSTLY ON HEATING FUNCTIONALITY, WITHOUT CONSIDERING TEMPERATURE ACCURACY OR HEAT DISTRIBUTION IN DETAIL.

MECHANISM EXPLORATION;

DURING THE DEVELOPMENT PROCESS, THE CONCEPT EXPANDED TO INCLUDE TEMPERATURE SENSING AND MORE CONTROLLED HEATING. THE IDEA OF USING A PROPELLER-BASED MIXING SYSTEM WAS EXPLORED TO IMPROVE HEAT DISTRIBUTION AND AVOID HOT SPOTS.
THIS STAGE INTRODUCED THE POSSIBILITY OF REAL-TIME TEMPERATURE MEASUREMENT INSIDE THE DRINK.

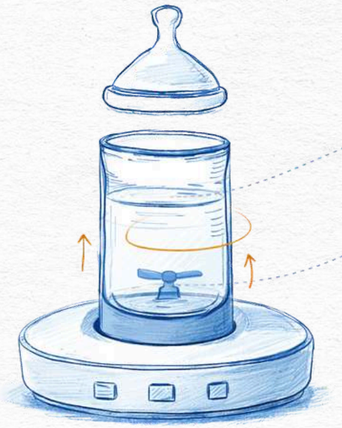
FORM DEVELOPMENT

THE PHYSICAL FORM OF THE PRODUCT EVOLVED DURING PROTOTYPING. THE EARLY CONCEPT BECAME BULKIER THAN EXPECTED, WHICH RAISED CONCERNS ABOUT USABILITY AND PRACTICALITY.
THIS LED TO REVISIONS IN THE PRODUCT STRUCTURE AND SCALE.

CURRENT DIRECTION

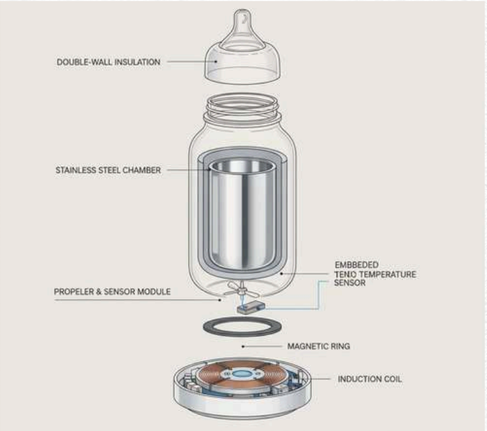
THE CONCEPT DEVELOPED INTO A SYSTEM THAT COMBINES CONTROLLED HEATING, TEMPERATURE SENSING, AND MIXING LOGIC TO PREPARE DRINKS AT A SAFE TEMPERATURE FOR BABIES AND YOUNG CHILDREN.
THE DESIGN NOW FOCUSES ON SAFETY, ACCURACY, AND EVERYDAY USABILITY.

How can the propeller inside the cup help parents feel more secure and calm during evening milk preparation?



- Eliminates need for manual stirring or checking temperature
- Continuous movement signals controlled, even preparation
- Reduces uncertainty, supports evening calm

Emotional reassurance



PROTOTYPE ITERATION

AI IDEATION → CONCEPT DEVELOPMENT → PROTOTYPE

ACCEPTED DECISIONS

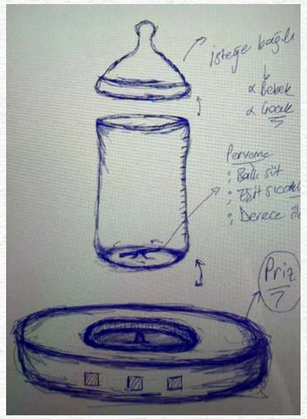
- REAL-TIME TEMPERATURE SENSING TO ENSURE DRINKS REACH A SAFE LEVEL.
- CONTROLLED HEATING SYSTEM TO AVOID OVERHEATING.
- PROPELLER-BASED MIXING LOGIC TO SUPPORT MORE EVEN HEAT DISTRIBUTION.
- PORTABLE PRODUCT STRUCTURE ALLOWING CAREGIVERS TO PREPARE DRINKS IN DIFFERENT ENVIRONMENTS.
- HANDLE INTEGRATION TO IMPROVE GRIP AND REDUCE THE RISK OF DROPPING THE CUP.

REJECTED DIRECTIONS

- HEATING-ONLY CONCEPTS, BECAUSE THEY DID NOT GUARANTEE SAFE DRINK TEMPERATURE.
- SOLUTIONS RELYING ON CAREGIVER ESTIMATION, SUCH AS TESTING TEMPERATURE MANUALLY.
- LARGE APPLIANCE-LIKE PRODUCT FORMS THAT REDUCED PORTABILITY AND PRACTICALITY.
- OVERLY COMPLEX MECHANISMS THAT COULD MAKE EVERYDAY USE DIFFICULT.

TRANSFORMED IDEAS

- THE CONCEPT SHIFTED FROM WARMING DRINKS TO DELIVERING DRINKS AT A SAFE TEMPERATURE.
- THE SYSTEM EVOLVED FROM SIMPLE HEATING TO CONTROLLED HEATING WITH TEMPERATURE FEEDBACK.
- THE PRODUCT DIRECTION MOVED FROM APPLIANCE-LIKE STRUCTURE TO A MORE USER-FRIENDLY PORTABLE SYSTEM.



FINAL DESIGN DECISION

FINAL DIRECTION STATEMENT

I WILL DESIGN A PORTABLE DRINK PREPARATION SYSTEM THAT HELPS CAREGIVERS PREPARE BEVERAGES FOR BABIES AND YOUNG CHILDREN AT A SAFE AND ACCURATE TEMPERATURE THROUGH CONTROLLED HEATING, REAL-TIME SENSING, AND MORE EVEN HEAT DISTRIBUTION.

WHAT THE DESIGN DOES;
THIS DESIGN FOCUSES ON SUPPORTING CAREGIVERS DURING DRINK PREPARATION BY REMOVING THE NEED TO ESTIMATE TEMPERATURE MANUALLY.
THE SYSTEM COMBINES CONTROLLED HEATING, TEMPERATURE SENSING, AND MIXING LOGIC TO HELP ACHIEVE A CONSISTENT AND CHILD-SAFE TEMPERATURE

REFLECTION;

DURING THIS PROJECT, AI HELPED ME EXPLORE IDEAS MORE QUICKLY AND ORGANIZE MY THINKING THROUGHOUT THE DESIGN PROCESS. IT WAS ESPECIALLY USEFUL DURING THE PROBLEM DEFINITION AND CONCEPT EXPLORATION STAGES, WHERE I USED IT TO CLARIFY THE DESIGN CHALLENGE AND GENERATE DIFFERENT POSSIBLE DIRECTIONS.

AI ALSO HELPED ME EVALUATE MY CONCEPT BY POINTING OUT POTENTIAL WEAKNESSES SUCH AS USABILITY ISSUES AND THE BULKY FORM OF THE PRODUCT. THIS ALLOWED ME TO RECONSIDER CERTAIN ASPECTS OF THE DESIGN AND REFINE THE CONCEPT TOWARD A MORE PRACTICAL AND USER-FOCUSED SOLUTION.

HOWEVER, THE FINAL DESIGN DECISIONS WERE BASED ON MY OWN EVALUATION OF USER NEEDS, SAFETY CONSIDERATIONS, AND THE PHYSICAL FEASIBILITY OF THE PRODUCT. AI SUPPORTED THE PROCESS BY EXPANDING POSSIBILITIES AND HELPING STRUCTURE MY THINKING, BUT THE DIRECTION AND PRIORITIES OF THE PROJECT WERE DETERMINED THROUGH MY OWN DESIGN JUDGEMENT.